

Project Proposal: Network Reliability under Edge Failures using Monte Carlo Simulation

Modern communication, transportation, and other infrastructure systems can be represented as graphs, where nodes stand for components and edges represent the connections between them. A common question in graph theory is how reliable these networks are if some connections fail randomly. This idea is called network reliability, which is basically the probability that the network is still connected after some edges stop working. This is important in real life because systems like the internet or power grids need to stay connected even when failures happen.

In this project, we will estimate network reliability using Monte Carlo simulation, which means we will run many random trials where edges fail and then check if the graph is still connected. This approach is useful because calculating the exact reliability is very hard for large graphs. Along with the simulation, we will also look at the **mathematical side** of the problem by studying concepts like **cut-edges (bridges)** and **minimum cuts**. We plan to include a few short proofs and explanations to show why certain graph structures are more fragile or more robust, and then compare those ideas with the simulation results on simple graphs like trees, cycles, and grids.

For the technical part, we will use **Python** for coding the simulations. We will use **NetworkX** to create and analyze graphs, **NumPy** for random sampling and calculations, and **Matplotlib** to draw graphs and charts of the results. The goal is to combine both math and coding so the project is not only theoretical but also experimental, showing how the theory matches what we observe in simulations.

References

1. Hadaj, P. (2023). Power grid failure simulations using PLANS and NetworkX combined with Monte Carlo algorithm. *Advances in IT and Electrical Engineering*, 29, 17–25.
2. Gertsbakh, I., Shpungin, Y., & Vaisman, R. (2014). *Network reliability Monte Carlo with nodes subject to failure*. *International Journal of Performability Engineering*, 10(2), 163–172.
3. MIT OpenCourseWare. *Mathematics for Computer Science – Graph Theory and Connectivity Lectures*.
4. StatQuest – Monte Carlo Simulation (YouTube)
<https://www.youtube.com/watch?v=7ESK5SaP-bc>